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**ELIGIBILITY**

Any Full Time Engineering Student / Faculty enrolled in Indian University.

CONTEST DATES

Contest Start Date :
01st July 2012

End Date for Submission of
Project Title-Abstract Form :
31st July 2012

End Date for Submission of
Final Project :
30th Sept 2012

Contest Winners to be
announced on
16th Oct 2012

For Registration and
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5 Use of multicore and parallel test systems

As the complexity and functionality of electronic devices grow exponentially (in sync with Moore's law), so does the cost of testing them. Minimising the cost of test can be challenging, but one way is to test more with less. The inherent parallelism that is made available by the graphical programming paradigm of software like LabVIEW from National Instruments and FlowStone DSP from DSP Robotics helps engineers immediately benefit from multicore processors and overcome the complexity associated with traditional text-based languages.

The trend of increasing clock speed to get better performance ended back in the early 2000s. Since then, processor manufacturers have implemented alternate technologies to ramp up performance while keeping the clock speeds around 3 GHz. These

technologies include the use of processors with multiple cores on a single chip, hyperthreading, wider buses and hyper transport. Moreover, the advancement of the process node to the current 22nm process by utilising 3D transistors has resulted in significantly faster, leaner and more efficient processors for use in embedded controllers and modular instrumentation.

Denver D'Souza, senior technical consultant at National Instruments India, says, "The reality that transistor density doubles every 18 months has led to significant advances in the performance of electronic devices. This is evident not only in the latest Intel Core i7 processors but in the shrinking of technology such as 64GB solidstate drives, which are now the size of a postage stamp. These technological advances translate into considerable cost reductions."

6 Merging of EDA tools and hardware test platforms

The extremely competitive environment in which electronics companies work now is shown by how next-generation communication protocols are barely labeled as standards before they can be seen in the market. For instance, the 802.11ac solutions have already been brought out by Broadcom even though it is yet to be ratified. In situations like these, companies go all out to get a jumpstart on the competition, and what better way to do this than to merge design and testing in order to accelerate the 'time to market'.

Adesh Jain, applications consultant at Agilent Technologies, explains why the traditional method is slow: "Traditionally, for any complete electronic product to be ready for the market, each component of the complete system is first

designed and verified with EDA tools, then prototypes are fabricated and tested, before the final product is released to the market. If discrepancies are found in the hardware at later stages, the whole cycle has to be repeated, which would result in loss of time as well as money for any organisation."

Proper verification at earlier stages reduces this time and effort to a great extent. The tests, specs, algorithms and plots used in the early stages of EDA are the same as measured on the test bench. The aim is to merge both the worlds and see if it is possible to save the design engineers' time by streamlining the flow and thus improve productivity while reducing the time to get the product out to the market.

The author is tech correspondent at EY Bengaluru